



Iron nitrides

Iron nitrides are inorganic chemical compounds of iron and nitrogen.

Chemical properties

Iron has five nitrides observed at ambient conditions: Fe_2N , Fe_3N_4 , Fe_4N , Fe_7N_3 and Fe_{16}N_2 . They are crystalline, metallic solids. Group 7 and group 8 transition metals form nitrides that decompose at relatively low temperatures—iron nitride, Fe_2N , decomposes with loss of molecular nitrogen at around 400 °C and formation of lower-nitrogen content iron nitrides. They are insoluble in water. At high pressure, stability and formation of new nitrogen-rich nitrides (N/Fe ratio at least one) were suggested^[1] and later discovered. These include the FeN , FeN_2 and FeN_4 solids which become thermodynamically stable above 17.7 GPa, 72 GPa and 106 GPa, respectively.^{[2][3][4][5][6]}

Health hazards

When heated to decomposition or exposed to humidity, iron nitride may emit toxic fumes of ammonia. It is considered a moderate explosion hazard. Inhalation of iron nitride dust or powder may cause irritation to the respiratory system and possibly acute iron poisoning or pneumoconiosis.

Research applications

Colloidal solution of magnetic iron nitride nanoparticles is a way to create ferrofluids.

Iron nitrides also make the strongest naturally magnetic material.^{[7][8][9]}

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